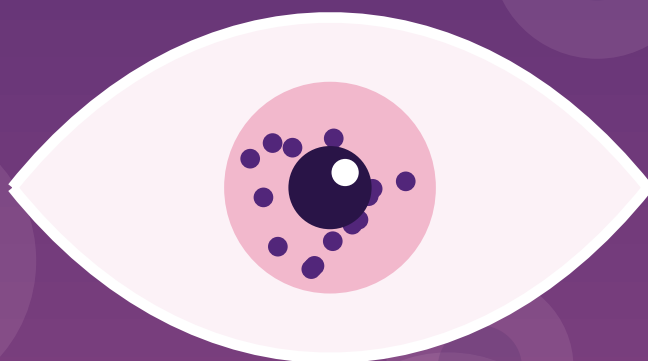




littleEye AI

User Guide for Physicians

Training and evaluation of AI models from histology slides
iPhone · iPad · Mac — everything directly on your device

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A project of three institutes of Comenius University in Bratislava · v1.0 (1) · August 2026

1 · What is littleEye AI

littleEye AI teaches artificial intelligence to look at histology slides through your eyes. You label micrographs with diagnoses — the app then trains a recognition model directly on your device, which then helps you evaluate new slides. Nothing ever leaves your iPhone, iPad, or Mac: no server, no internet connection, no data collection.

What's in the app

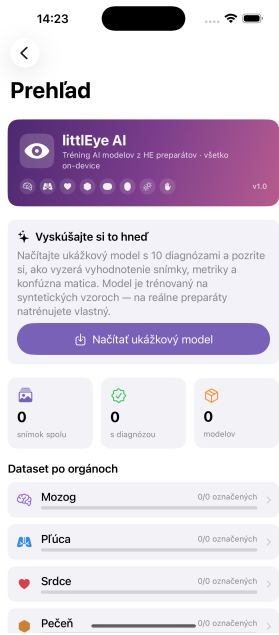
- Dataset — H&E micrographs of eight organs (brain, lung, heart, liver, kidney, spleen, pancreas, skin)
- Slide review — colleagues assign diagnoses; every assessment is saved with a name (audit trail)
- Training — your own Core ML model with honest metrics (test set, confusion matrix)
- Image evaluation — a leading diagnosis with a confidence percentage and a draft finding text
- Cases — the autopsy workflow: slides + case history + autopsy conclusion + toxicology → a concise summary
- Diagnosis atlas and Education — descriptions of ~40 diagnoses, special stains, IHC, cellular elements, pigments
- le3AI model files — portable files for sharing trained models between colleagues

Requirements

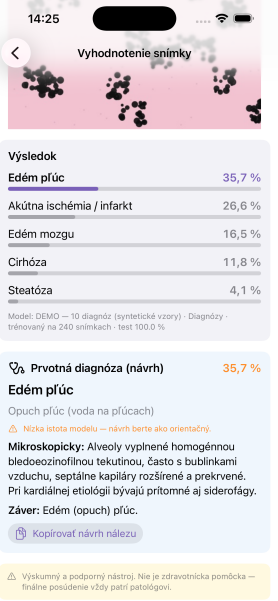
iPhone (iOS 17+), iPad (iPadOS 17+; training works best on iPads with an M-series chip), Mac (macOS 14+, Apple Silicon). The app is bilingual (SK/EN) with both professional and lay terminology — switch languages using the globe icon.

2 · First launch — try it in 3 minutes

1. Open the app, choose your language, and enter your name (it's saved with assessments and models — audit trail).
2. On the Overview screen, tap Load Sample Model — a bundled demo model with 10 diagnoses.
3. The app opens Image Evaluation. Pick any photo and see what the result looks like: diagnoses ranked by confidence in %, a leading diagnosis with a microscopic description, and a draft conclusion.



Overview, with the sample model card



Evaluation: confidence levels, leading diagnosis, draft finding

Important

The sample model is trained on synthetic patterns — it will not recognize real specimens. It's meant only to help you get familiar with the app. The application gains its real power only once you train a model on your own images.

3 · Image capture — the basis of everything

The model will only ever be as good as your images. Six golden rules:

Six golden rules of image capture

- One magnification per model — we recommend 100× or 200×; the magnification is recorded on import
- The finding should fill most of the field of view — the model evaluates the whole image
- Sharp, evenly exposed images with correct white balance
- Avoid section edges, tissue folds, overlaps, and staining artifacts
- Take several images per slide from different representative areas
- “Normal” is a full-fledged class — the model needs it just as much as pathological ones

How to get images into the app

- Microscope camera / slide scanner — export JPEG/TIFF and transfer via Files (USB-C drive, network share, iCloud) or AirDrop; the app needs no drivers
- iPhone with an eyepiece adapter — import photos directly from Photos
- In the dataset section, choose an organ, select the magnification in the toolbar, and tap From Photos or From Files

4 · Labeling and review with colleagues

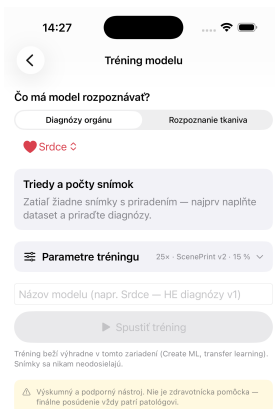
1. Tap an image to open its detail view — choose a diagnosis from the list (you can extend the list with your own entries).
2. In the Slide Review section, colleagues go through the images one by one and assign diagnoses. Every vote is saved with a name; the final diagnosis is decided by majority.
3. Press and hold a diagnosis to see its Atlas description, so everyone labels by the same criteria.
4. Marking a region: if an image contains more than one finding, tap Mark Region (crop) in the detail view, drag to outline the area, and save it as a separate training image. A crop is how you tell the model WHERE to look.

Example 1 — a mini pulmonary model

Dr. Eye wants to build a model for three pulmonary diagnoses. At 100× magnification, they collect: 35 images of pulmonary edema, 32 of bronchopneumonia, and 30 of normal tissue — from twelve different autopsies. They and a colleague cross-check the images in the Review section; for four disputed cases, a third vote decides. The dataset is ready for training.

5 · Training a model

1. Open Training, choose Organ Diagnoses and an organ (or Tissue Recognition for a model that distinguishes organs).
2. Check the class list — the minimum is 2 classes with 5 images each; we recommend 30+ per class. The app will flag class imbalance as well as mixed magnifications.
3. In Training Parameters, leave the default values (25 iterations, ScenePrint v2, 15% test split).
4. Tap Start Training. On an M1 iPad, training a few hundred images typically takes 1–3 minutes; you can watch its progress and cancel it.



How to read the results

- Training / validation / test — the percentage of correctly classified images; the test result matters most (images the model never saw while learning)
- Confusion matrix — row = true diagnosis, column = prediction; numbers off the diagonal show what the model confuses
- Sensitivity and precision for each class individually

A large gap between training and test accuracy means overfitting — add more varied images.

Example 2 — reading the confusion matrix

The model from Example 1 reaches 92% on the test set. The confusion matrix shows it labeled 3 bronchopneumonia images as edema — both classes have congested septa. Fix: add bronchopneumonia images with pronounced purulent exudate (crops from areas rich in neutrophils) and retrain the model. After adding them, the test accuracy rises to 97%.

6 · Evaluating an image in practice

1. Open Image Evaluation, select a model at the top, and add an image (Photos or Files).
2. Result: diagnoses ranked by confidence in %. Confidence is always part of the result — never a bare statement.
3. For the most confident class, a Leading Diagnosis appears — a microscopic description and a draft conclusion from the Atlas. Use the Copy Draft Finding button to transfer the ready-made text into your report (Word — Cmd/Ctrl+V).

Principle

Below 80% confidence, the app itself flags the suggestion as indicative only. AI is a second opinion, not a decision — the final assessment always rests with the pathologist or forensic physician.

7 · Cases — the autopsy workflow

1. In the Cases section, create a case using the protocol number (e.g., 680/2026).
2. In the toolbar, choose the organ and magnification, then import slides from the scanner; you can review each one with zoom (pinch, double-tap), and assign a diagnosis and description.
3. Add the case history, post-mortem conclusion, and toxicology — brief is fine, just the key facts.
4. Tap Evaluate Case. The app assesses all slides (your diagnoses take priority; unassigned ones are judged by the model) and generates a concise summary.
5. Copy (to Word) — paste the summary into your report. Then archive the case.

Example 3 — sample summary (abridged)

CASE 680/2026 — littleEye AI summary
 FINDINGS FROM SLIDES (n = 7)

- Lungs (3): pulmonary edema; venous congestion (model, 91%).
- Heart (2): acute ischemia / infarction.
- Liver (2): steatosis (model, 88%).

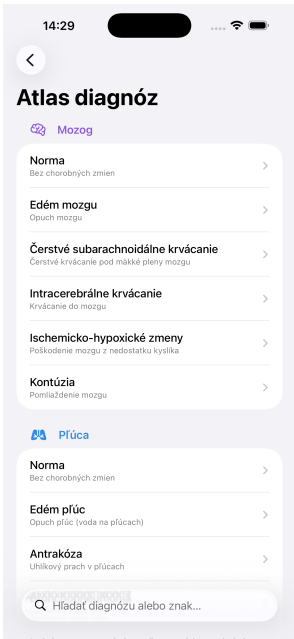
CONCLUSION — from slides only
 1. Acute ischemia / infarction. 2. Pulmonary edema, venous congestion. 3. Steatosis.

CONCLUSION — with case data

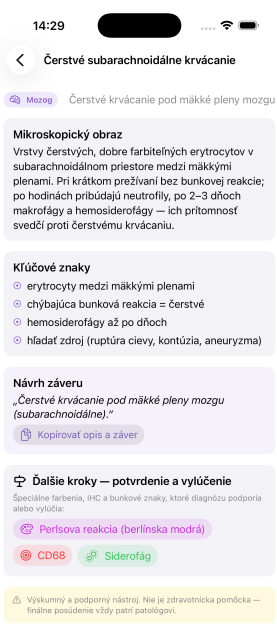
- Cardiac history supports the finding of acute ischemia.
- History of alcohol use supports the finding of hepatic steatosis.

8 · Sharing models and backups

- Model export: Models & Export → Export .le3AI → save it and send it to a colleague (AirDrop, email, Files). The file carries the model, metrics, confusion matrix, and the authors' names — never the specimen images.
- Import: a colleague opens the file with a tap/double-click or via Import .le3AI; a checksum guards its integrity.
- Archive backup: Cases → Settings → Back Up Archive to Disk — a complete backup (registry, images, models) to an external disk or network storage.



Diagnosis atlas — consistent criteria for the whole team



Diagnosis detail: description, key features, conclusion, next steps (stains/IHC)

9 · Vision: a shared “brain” for institutes and universities

Every physician who labels specimens in littlEye AI builds a small piece of collective experience. The .le3AI format and the backup system were designed from the start so these pieces can be combined — a personal tool can gradually grow into a shared diagnostic memory for the whole field.

PHASE 1 — PHYSICIAN today	PHASE 2 — INSTITUTION next step	PHASE 3 — NETWORK goal
Everyone trains locally on their own specimens. Models are shared as .le3AI files — a colleague in Košice emails a model to a colleague in Bratislava.	The institute appoints a curator. A shared drive holds a collection of vetted models — one “champion” per organ, validated on images from a reviewer other than the one who trained it. The confusion matrices and audit trail embedded in .le3AI make quality auditable.	The Forensika server: a registry of models and anonymized datasets. Institutes and faculties act as nodes — contributing annotations, downloading shared models; disputed specimens are resolved by consensus across several institutes. The app’s backup format is already built for this exchange today.

Principles the shared brain rests on

- Consensus instead of authority — the final label is set by majority vote; disagreements are the most valuable teaching material
- Cross-institutional validation — tested on images from an institute that did not train it
- Audit trail — for every model you can trace who trained it, when, on how many images, and with what accuracy
- Anonymization — only micrographs without identifying data are ever shared; models themselves never contain images at all
- Teaching — students learn from specimens with a consensus diagnosis; the Atlas and Education sections serve as course materials right inside the app

Why this matters

Today, a pathologist’s experience retires along with them. A labeled specimen is experience that stays behind: it can be shared, measured, challenged, and improved. A network of institutes exchanging models and annotations builds something no textbook can offer — a living, verifiable, ever-growing diagnostic memory for the field.

10 · Usage principles and limitations

- littlEye AI is a research and support tool — not a medical device; it does not provide a diagnosis
- The model’s output is always only a second opinion with a stated confidence level; the final assessment rests with the pathologist or forensic physician
- A model trained at one magnification and stain may not work under other conditions
- Before practical deployment, validate the model on an independent image set
- The user is responsible for ensuring images comply with institutional policies and patient data protection

About the project

littleEye AI is a joint project of three institutes of the Comenius University Faculty of Medicine in Bratislava: the Institute of Histology and Embryology, the Institute of Pathological Anatomy, and the Institute of Forensic Medicine and Forensic Toxicology. The content is developed in collaboration by Dr. Ján Šikuta, MD, and other members of these institutes.

The app and its ideas took shape over a long time — the content matured and was refined based on experience with the 4n6 Tools and Colorime3ka apps (both available on the App Store).

Contact & support

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Detailed guides are available directly in the app: Methodology for Trainers, Hardware & Imaging, Atlas, Education.